TRACE-Q: An Exploratory Multi-Agent Toolkit for Evidence-Grounded Review of Quantitative Research Reporting

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Abstract

Quantitative reporting issues can be difficult to assess when relevant evidence is distributed across an article, supplements, protocols, and statistical analysis plans. We describe TRACE-Q, an exploratory multi-agent toolkit that organizes source mapping, quantitative checks, inclusion screening, evidence rechecking, and audit into a documented workflow. Version 1.6.2 preserves source locations, intermediate records, and candidate histories while reserving final judgments for human review. We examined its application to an availability sample of 44 randomized trial packages published in *JAMA* in 2018, 2019, 2024, and 2025, using completed human adjudication workbooks as the final results. Of 440 candidates, 413 were retained and 27 excluded. Retention included descriptive protocol–report differences and did not imply confirmation of a reporting issue. The principal temporal endpoint comprised 41 confirmed or probable substantive statistical issues in report materials, occurring in 22 of 44 papers. Mean counts were 1.05 per paper in 2018–2019 and 0.84 in 2024–2025; the later-minus-earlier difference was -0.21 (95% bootstrap confidence interval, -1.02 to 0.56; exact paper-level permutation $P = 0.654$). The findings illustrate how evidence-linked candidates can be organized for human assessment. They do not establish diagnostic accuracy, improved efficiency, or a temporal change in reporting quality. Missed issues were not assessed, and the availability sample and single recorded human reviewer limit generalization. TRACE-Q should be regarded as an exploratory aid for locating and examining reporting issues.

Keywords: quantitative reporting; large language models; multi-agent workflow; human adjudication; randomized trials; evidence synthesis

1 Introduction

Quantitative research reports distribute information across the main article, tables, figures, supplementary material, protocols, and statistical analysis plans. Interpreting a result may therefore require a reader to reconcile several descriptions of its population, outcome, analysis, and uncertainty. Disagreements can complicate interpretation and the extraction of data for subsequent research. In a reproducibility study of systematic reviews of adverse events, Xu et al. found that extraction errors sometimes changed the direction or statistical significance of pooled effects, illustrating why the connection between a reported number and its source matters.[1]

Large language models (LLMs) have been evaluated for several tasks relevant to medical and quantitative research. Ruta et al. examined data management and univariate statistical analysis using ChatGPT and found that inferential results depended on the specificity of the instructions.[2] Graf et al. evaluated reference generation for obstetric trials and documented frequent inaccurate citations.[3] In ophthalmology,

Paris et al. compared LLM triage of hypothetical oculoplastic referrals with specialist assessments and found limited exact agreement.[4] These applications demonstrate the range of tasks that can be presented to an LLM, while emphasizing the need to evaluate the particular task and to verify its outputs.

Recent work in computer science has examined the more specific task of finding problems within documents. These arXiv preprints use different evaluation targets: FIND asks models to identify expert-inserted inconsistencies in long documents and provide supporting passages; FLAWS tests the identification and localization of inserted errors that undermine scientific claims; and SPOT evaluates the recovery of documented scientific errors associated with corrections or retractions.[5–7] A separate preprint by Bianchi et al. applied an LLM checker to published artificial intelligence papers and obtained human assessments of a sample of its findings.[8] Together, these studies distinguish two questions: whether a system can recover known errors and whether its newly proposed findings withstand scrutiny. Their results cannot be treated as interchangeable estimates of performance because the documents, error definitions, reference standards, and evaluation procedures differ.

Medical studies have also considered manuscript checking directly. La, Paris, and colleagues evaluated four LLMs using 40 ophthalmic articles, including 30 with one planted inconsistency each. They assessed detection, localization, explanation, and correction, and manually reviewed additional findings.[9] Dong et al. evaluated a structured prompt for reviewing selected categorical analyses in medical papers. Their study identified useful checking capabilities alongside extraction failures, omitted analyses, and dependence on the reported values being checked.[10] These findings motivate supplementary review, but they also show why a final list of concerns is insufficient: a reviewer needs to establish what was examined, which source supports each concern, and how the conclusion was reached.

A practical workflow must preserve this information when review spans multiple files or working contexts. Reusable text and page images can support later inspection, provided that derived material remains linked to the original source. A persistent record of source coverage, numerical relationships, candidate findings, and screening decisions allows later inspection after an agent's immediate context changes. Existing agent environments provide file access, document tools, task delegation, and persistent artifacts through which such a workflow can be implemented. These capabilities offer an implementation route; their presence alone does not establish greater accuracy or lower review cost. Human responsibility and transparent reporting of AI use remain necessary in medical publication.[11]

We present TRACE-Q, an exploratory multi-agent toolkit for evidence-grounded review of quantitative research reporting. We describe Workflow 1.6.2, including source mapping, focused checks, candidate screening, evidence rechecking, and report generation. We then examine human adjudications of candidate findings in an availability sample of 44 randomized clinical trials published in JAMA in 2018–2019 or 2024–2025. Our objectives are to make the review process explicit, describe the findings retained after human assessment, and explore differences between the two publication periods. The application does not provide an exhaustive reference set of all reporting problems and therefore does not estimate sensitivity for undiscovered issues. The period comparison is descriptive and does not establish the causes of any observed difference.

2 The TRACE-Q Framework

TRACE-Q organizes the review of a supplied paper package into source mapping, quantitative checks, inclusion screening, evidence rechecking, and reporting. Here we describe the specification for workflow version 1.6.2. Its purpose is to identify candidate reporting issues that could affect interpretation of a reported result or extraction of a value for evidence synthesis, and to prepare evidence for human adjudication. The review uses the article and its supplied supplements, protocol, statistical analysis plan (SAP),

and structured files. It does not establish validity from participant-level data or adjudicate the conduct of a study. Figure 1 summarizes the sequence.

2.1 Source Coverage and Durable Evidence

A coordinator assigns bounded tasks to specialist agents and maintains durable artifacts linking each stage to its evidence. These include source and extraction inventories, coverage manifests, quantitative relationship inventories, checker findings, screening decisions, a candidate ledger, recheck records, and an audit. Specialists write complete results to assigned files and return short status messages. This design makes evidence available beyond an individual agent's conversational context and makes incomplete assignments visible to subsequent stages.

The source inventory records document identity, location, role, and unit count. Units correspond to the material supplied, such as PDF pages, workbook sheets, or paragraph and table groups. Existing native text, layout text, optical character recognition (OCR) output, rendered pages, and table extractions may be reused after their coverage and usability are assessed. Missing, stale, or partial derivatives generate explicit assignments for direct-source mapping. For every source, the numbers of reusable units and units requiring fresh mapping must sum to the total. All units must be mapped before completion. This coverage requirement cannot guarantee that every relevant relationship has been correctly recognized.

Reuse applies to extraction assets, not previous candidate selections or classifications. Discovery starts again from the supplied evidence. Source files and reused assets are hashed before review and checked again at completion. Derived text serves as a locator and transcription aid: a candidate must ultimately refer to an exact PDF page, worksheet and cell range, CSV record, or document paragraph or table. A discrepancy introduced only by OCR or extraction is recorded as a transcription matter.

Mapping of the main article and supporting documents proceeds concurrently over nonoverlapping assignments. Mappers record counts, denominators, estimates, intervals, test statistics, and the labels needed to interpret them, including population, time point, comparison, reference group, model, and adjustment. Numeric relationships receive stable `N` identifiers and inferential relationships receive `S` identifiers. The configured bounds are 200,000 normalized text characters, 32 rendered pages, or 120 relationships per shard, and 60 candidates per recheck shard; up to six specialists may run concurrently. These bounds partition work rather than limit discovery. No candidate count target or early stopping rule is specified.

2.2 Review Domains and Document Relationships

Checks cover counts and denominators, estimate and test logic, statistical method logic, and planned versus reported statistical design. Each candidate receives exactly one of four review domains (Table 1). Statistical method checks address source-grounded contradictions or gaps that prevent identification of the analysis producing a reported number. They do not substitute a preferred analysis for an explicitly stated, internally coherent method.

Document roles depend on content. The article and results supplements are `REPORT`; protocols, SAPs, amendments, and prospective operational documents are `PLAN`, even when packaged as supplements. A conflict between report and plan is `P_R`; one confined to plan material is `P_P`; one between different report documents is `S_C`; and one within a report document is `S_W`. All applicable relationships are recorded. The primary type follows the precedence order $\texttt{P_R} > \texttt{P_P} > \texttt{S_C} > \texttt{S_W}$. A document that merely corroborates one side does not create an additional relationship. The primary type determines the domain for plan-related candidates; report-only candidates are assigned by content. A protocol–report classification does not itself establish a protocol violation.

Table 1: The four review domains in workflow 1.6.2. Relationship codes describe the locations in conflict, rather than severity.

Review domain	Relationships	Scope
Statistical results	`S_C`, `S_W`	Counts, denominators, estimates, intervals, tests, and statistical method logic in report materials.
Report internal inconsistency	`S_C`, `S_W`	Conflicting attribution of a reported number to an outcome, time point, analysis set, or comparison, leaving unclear which result applies.
Protocol–report	`P_R`	Differences between report and plan in statistical design or analysis elements.
Plan material	`P_P`	Inconsistencies within or between plan documents in statistical design elements on which the reported analysis relies.

2.3 Checks and Inclusion Rules

After mapping, numeric, cross-source, and first-pass statistical checkers work concurrently. Comparisons match population, outcome, time point, contrast, and model before assessing agreement. Checks include participant-flow arithmetic, estimate containment within intervals, direction and reference-group agreement, compatibility of intervals and tests, stated decision margins and multiplicity rules, sample-size calculations, and planned versus reported outcomes and analyses. Model-based estimates are not assessed against crude arithmetic unless identified as crude or unadjusted. Diagnostic approximations are labeled and do not replace the reported analysis.

Registration requires reproduction of the observation and fulfillment of the criteria for either Route A or Route B. Route A comprises five categories that must always be registered: interval–test or interval–P-value incompatibility, including a point estimate outside its interval; printed value slips that change a reading; counts, denominators, and percentages that fail to reconcile within one display; a result not reproduced by a method that reproduces other rows in that display; and specified changes or omissions in planned outcomes or analyses. The last category includes changed outcome status or composite definition, an analysis described as prespecified but absent from the supplied plan, and an unreported planned primary or key secondary analysis. Reproduced Route A findings do not additionally need to satisfy Route B's context criterion.

Route B requires all three conditions: the observation is within scope, its intended meaning cannot be resolved immediately by a statistically literate reader in the relevant field, and it could affect interpretation or extraction of a named result. Cross-location differences in peripheral material, such as copied baseline characteristics or administrative details, are generally screened out unless they affect a primary or key secondary result, analysis-population size, or planned decision rule. Wording differences, unspecified details without a conflicting statement, and preferences for another method do not qualify on their own.

Numeric reproduction uses a specified display rule. Each displayed input is assigned its half-unit rounding interval, and these intervals are propagated through the calculation. Let I_r denote the resulting interval and I_p the interval around the printed comparator. Their gap g is zero if they overlap; otherwise it is their minimum separation. The residual relative discrepancy is $g/|x_p|$, where x_p is the printed comparator or the designated reference when comparing two printed copies. A zero gap or relative discrepancy at most 0.05 is screened out; a discrepancy above 0.05, or any positive gap when the comparator is zero, is reproducible under this rule. The recomputed interval and discrepancy are retained.

The 5% cutoff is an operational screening heuristic beyond ordinary rounding intervals, not a statistical significance tolerance or a demonstrated boundary of substantive importance. It can exclude numerical

differences that exceed rounding alone. Route A's interval–P-value/test category is exempt: confidence and significance levels follow the source, with 95% and 0.05 used when unspecified. If the source attributes the interval and test to different analyses, this is retained as an alternative interpretation for adjudication rather than used to suppress the candidate. A display-zero P value alone does not qualify.

2.4 Independent Screening, Rechecking, and Audit

Each checker records qualifying findings with checker-specific identifiers and logs apparent issues it excludes using a reason code. A fresh inclusion screener rereads cited locations and independently applies the criteria to every finding. Repeated manifestations of the same rule and evident cause are consolidated while retaining all affected locations. Included issues receive stable `C` identifiers linked to their contributing findings. Screened-out observations remain traceable in the screening record without receiving candidate identifiers.

A separate evidence rechecker examines every registered candidate against the direct source, including the printed values, comparator, rule, calculation, missing inputs, and source-grounded alternative interpretations. It distinguishes direct observations from inferred explanations and confirms document roles and classification. Evidence is recorded as `reproduced`, `reproduced_after_transcription_correction`, or `not_reproduced`. Even the last state retains the original candidate identifier and card, preserving the history rather than silently removing a registered item.

A second statistical pass uses a different fresh agent after the ledger, screening log, and recheck are available. It revisits every statistical relationship and checker exclusion, including checks that produced no candidate. Because it reads prior results, this is a staged reassessment rather than a blinded replication. New findings undergo screening by another fresh screener, registration without renumbering existing candidates, and source rechecking. The final auditor examines coverage, decisions to include or exclude findings, classification, evidence states, and execution records. Inclusion is marked `CONFIRMED` or `QUESTIONED`, with reasons; questioned candidates remain visible.

2.5 Report and Human Adjudication

The report contains every stable candidate, its source links, compared values, reasoning, calculation, alternative interpretations, and remaining uncertainty. Counts reconcile across the ledger, recheck, audit, and report. Mechanical validation checks coverage records, identifier agreement, classification fields, evidence links, execution metadata, and source integrity. A successful structural check does not establish scientific correctness. Markdown and standalone HTML outputs prominently state that human adjudication is pending; adjudication is recorded separately in a review workbook.

The supplied implementation specifies `gpt-6.1-sol` as the default single model across roles, with higher reasoning effort for statistical review, screening, rechecking, and audit; the actual run model and agent identities are recorded. Fresh agents separate stage contexts but do not ensure independent judgments. The architecture therefore documents how candidates arose and what evidence supports them, while leaving their final disposition to human reviewers. Reports assign neither severity nor a recommended disposition, and potential downstream effects remain conditional on confirmation. Zero candidates is an acceptable result after complete documented coverage.

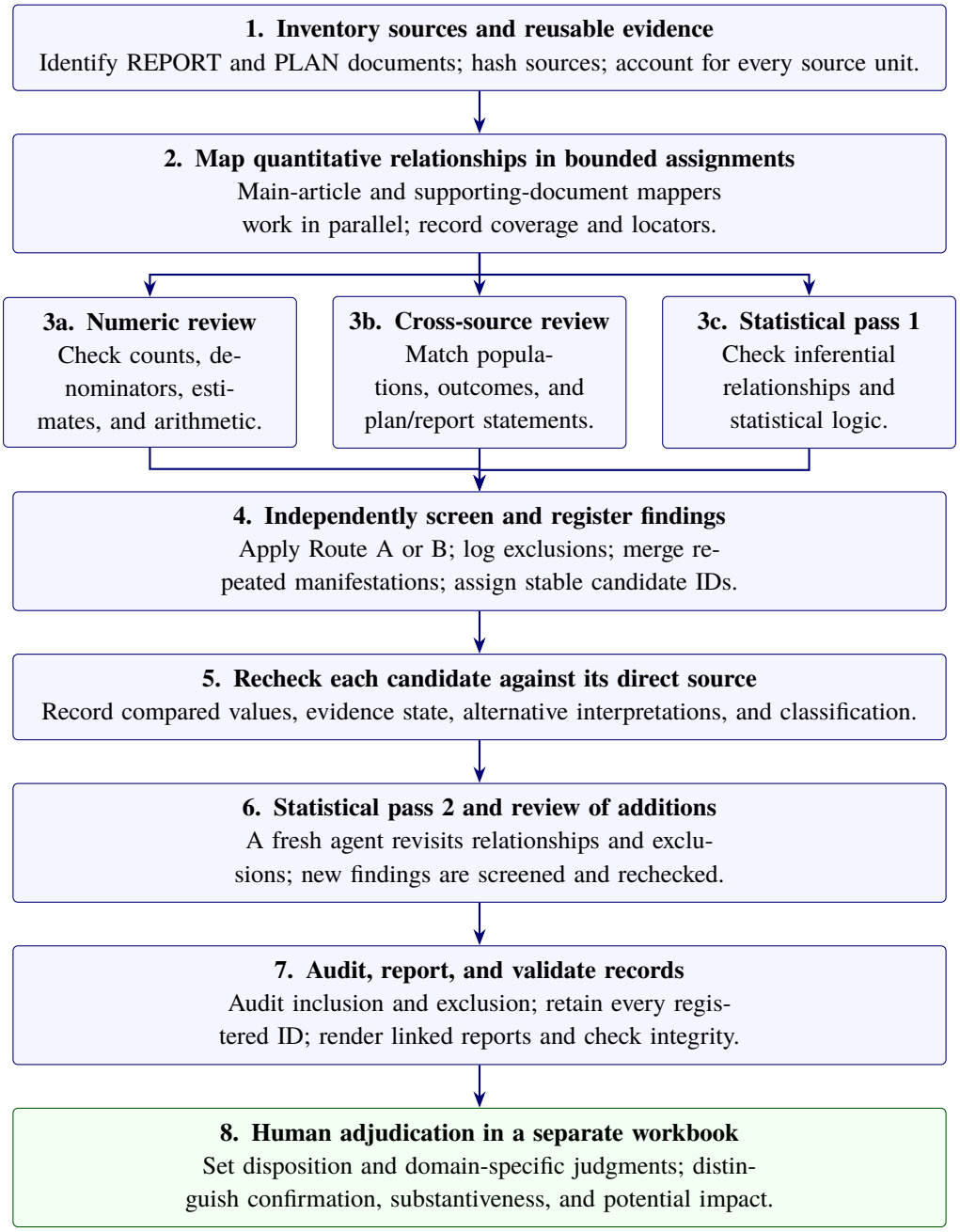


Figure 1: TRACE-Q workflow 1.6.2 and its handoff to human adjudication. Each stage writes durable records that can be examined by later stages. Excluded observations and questioned candidates remain traceable; additional findings enter screening and source rechecking before report completion. Automated evidence checks and audit decisions do not determine the final human disposition.

3 Application to Selected JAMA Trial Reports

3.1 Paper Selection and Source Packages

We examined an availability sample of 44 original randomized clinical trial reports published in *JAMA* in 2018, 2019, 2024, and 2025. The working eligibility criteria required a primary randomized trial report, publication in one of these four years, and availability of the main article and accompanying documents for local review. Papers were selected according to availability rather than probability sampling. The final workbooks contained 10, 9, 10, and 15 papers in the respective years. We used the project's online-first year assignments; a DOI containing an earlier year did not change that assignment. A nonrandomized report in the earlier working collection, `jama.2019.14231`, was excluded from the trial cohort and was absent from the final workbooks. We did not have a complete screening log for all potentially eligible journal articles and therefore could not estimate a journal-wide inclusion fraction.

A paper package comprised the main report and its locally available supplementary documents, including protocols, amendments, SAPs, and supplementary results when supplied. An SAP was not necessarily a separate file, and absence of a separate SAP did not imply absence of planning information. Document roles were determined from content, as described in Section 2.2. The 44 local packages contained 188 PDFs with 5,559 physical pages and one supplementary Excel file with one worksheet. These counts describe the available files, including references and administrative material, rather than the number of independent quantitative checks. Table 2 summarizes the packages by year.

3.2 Workflow and Adjudication Provenance

The candidate catalogues linked every workbook row to a workflow 1.6.2 report through a stable key consisting of year, paper identifier, and candidate identifier. The workbook version, 2.4.1, refers to the adjudication form and its codebook; it is distinct from the workflow version. We matched all 440 keys to the source catalogues and verified the stored SHA-256 hashes against the corresponding final reports. Recorded runs began between October 11 and October 14, 2026, and the routing records named `gpt-6.1-sol`. The supplied implementation used Codex, with reasoning effort assigned by role. The historical Codex client version was not recovered in this analysis.

We treated the four completed Excel files supplied for this study as the final human review results. They recorded one reviewer identity, `R_Bulun`; no second set of independent human ratings was available for agreement analysis. We used the entered decisions without substituting the workflow's evidence states or internal audit judgments. An automated recheck marked as reproduced or an inclusion audit marked as confirmed does not constitute human assessment of the scientific issue.

3.3 Human Review Codebook and Analysis Eligibility

The codebook separates disposition from subsequent domain-specific judgments. For every candidate, the reviewer chooses retain, exclude, or discuss and records completion. Retain keeps a candidate in the applicable results; exclude removes it from the retained findings, for example when the apparent discrepancy has a satisfactory explanation; discuss defers a decision. A blank indicates an unanswered item rather than a negative judgment. Every supplied candidate was marked `primary_done`, and none had a blank or deferred disposition.

For retained statistical candidates, two judgments are required: whether the issue is confirmed, probable, absent, or indeterminate; and whether it is substantive, limited to presentation, or indeterminate in this respect. Confirmation concerns evidential support, whereas substantiveness concerns possible effects on

interpretation or data use. Thus, a confirmed issue can be a minor presentation problem. An optional downstream-impact field records direct, possible, unlikely, or unknown implications for later extraction or analysis. Direct impact means the affected quantity may itself be extracted; it does not document harm or a change in meta-analysis results.

Retained report-internal and planning-material candidates receive an assessment of substantive inconsistency, minor presentation inconsistency, no inconsistency, or indeterminate status. For protocol–report candidates, an optional interpretation describes a substantive difference, minor presentation difference, version or timing ambiguity, no inconsistency, or an indeterminate comparison. A substantive protocol–report difference does not establish an unjustified change. These interpretations are descriptive and are excluded from temporal comparisons, as are planning-material findings.

We defined the principal temporal endpoint as the number of retained statistical candidates per paper that were confirmed or probable and judged substantive. The secondary endpoint counted retained substantive report-internal inconsistencies. Both required an original report-only relationship, `S_C` or `S_W`. Domain changes made during human review were preserved in descriptive summaries, while the original relationship code remained fixed for temporal eligibility. This rule prevented an original plan-related comparison from entering the temporal analysis solely because its domain changed during review. We also reported the proportion of papers with at least one eligible issue. Sensitivity analyses restricted statistical issues to confirmed cases and combined the principal and secondary endpoints. These analyses describe the supplied final results; we did not assume that the analysis had been prospectively registered.

3.4 Statistical Analysis

We summarized candidates and dispositions by year and human-assigned domain. Paper-level counts retained all 44 papers, including those with no eligible substantive issue. The early period comprised 19 papers from 2018–2019, and the later period comprised 25 from 2024–2025. The paper was the unit of analysis because candidates within a paper share documents, analysis methods, and review context.

For a count endpoint Y_i, the comparison statistic was the difference in mean counts, later minus earlier,

$$T = \frac{1}{25}\sum_{i \in L} Y_i - \frac{1}{19}\sum_{i \in E} Y_i.$$

We calculated a two-sided exact permutation P value by reallocating the 44 paper-level counts to groups of 19 and 25 and summing the probabilities of statistics with absolute value at least as large as the observed value. A dynamic programming calculation counted allocations by the size and integer sum of the early-period subset, avoiding explicit enumeration of all $\binom{44}{19}$ assignments. Ties were included. For the paper-level binary outcome, the same calculation used $\mathbb{1}(Y_i > 0)$, giving a test of the difference in observed proportions. Candidate labels were never permuted separately from their papers.

We calculated 95% percentile bootstrap confidence intervals for differences using 20,000 resamples of papers within each period, with random seed 20261022. All outcomes from a resampled paper stayed together. A Clopper–Pearson interval summarized the overall proportion of papers with at least one issue meeting the principal endpoint. These intervals describe resampling or binomial uncertainty conditional on the selected papers and do not account for availability sampling. Exact permutation calibration requires exchangeability under the null; differences in trial mix, package content, or review history may violate that assumption. Accordingly, all comparisons were exploratory, P values were unadjusted, and no confirmatory significance threshold was applied. A large P value was not interpreted as equivalence. Analyses used R 4.5.0 and `readxl` 1.5.0; scripts, derived tables, input hashes, and session information accompany the draft.

Table 2: Paper packages and final human dispositions by publication year.

Year	Papers	PDFs	Pages	Candidates	Retained	Excluded
2018	10	37	1070	115	107	8
2019	9	42	994	79	76	3
2024	10	41	1097	113	107	6
2025	15	68	2398	133	123	10
Total	44	188	5559	440	413	27

The 44 packages were selected by availability. PDF page counts exclude one supplementary XLSX file (one worksheet) in the 2025 cohort. Year is the recorded publication cohort and need not match the DOI year. Retention does not imply that every candidate is a confirmed reporting issue.

4 Results

4.1 Candidate Disposition and Human Judgments

The 44 paper packages contributed 440 candidates: 115 from 2018, 79 from 2019, 113 from 2024, and 133 from 2025 (Table 2). Human review retained 413 (93.9%) and excluded 27 (6.1%). There were no deferred dispositions or incomplete candidate rows. These counts begin with candidates already registered by the workflow; observations screened out before registration were not part of the human workbook denominator.

The final human-assigned domains comprised 173 statistical candidates, 12 report-internal candidates, 216 protocol–report candidates, and 39 planning-material candidates (Table 3). Exclusions comprised 18 statistical, 2 protocol–report, and 7 planning-material candidates. Five domain assignments changed during human review. The disposition totals and domain changes were retained exactly as entered.

Among the 155 retained statistical candidates, 139 were confirmed, 15 probable, and 1 indeterminate. Of the confirmed candidates, 37 were substantive and 102 were limited to presentation. Of the probable candidates, 4 were substantive, 9 were limited to presentation, and 2 were indeterminate in this respect. The candidate with indeterminate confirmation status was classified as a minor presentation matter. The optional impact judgments were direct in 29 cases, possible in 40, unlikely in 85, and unknown in 1. These judgments describe potential downstream use rather than measured consequences.

All 12 report-internal candidates were retained: 5 were substantive, 6 minor presentation inconsistencies, and 1 indeterminate. One substantive candidate had an original `P_R` relationship and therefore remained outside the temporal endpoint. Among 32 retained planning-material candidates, 4 were substantive, 25 minor, and 3 indeterminate. The 214 retained protocol–report candidates comprised 179 substantive differences, 14 minor presentation differences, 6 version or timing ambiguities, and 10 indeterminate comparisons; 5 had no optional interpretation. These 5 blanks were permissible under the codebook and were not treated as negative findings.

Three adjudicated examples illustrate these distinctions. For `jama.2018.14280`, candidate C001 concerned the identity of a reported binary risk ratio and was retained as a confirmed, substantive statistical issue with direct potential impact on extraction. For `jama.2017.21906`, C001 compared a planned medication composite containing statin use with the reported composite, which omitted statin use; the report disclosed that those data had not been collected. It was retained as a substantive protocol–report difference, without an inference that the change was unjustified. Candidate C002 for `jama.2018.0156` was excluded because the review found that the secondary-outcome and safety-analysis descriptions could coexist within the supplied plans. These examples summarize the final adjudications and do not supply corrected values.

Table 3: Candidate dispositions using the final human review domains.

Review domain	Candidates	Retained	Excluded	Retained (%)
Statistical results	173	155	18	89.6
Report internal inconsistency	12	12	0	100.0
Protocol–report	216	214	2	99.1
Planning material	39	32	7	82.1

Five candidates changed domain during human review. Protocol–report differences and planning-material findings are descriptive; they do not enter the temporal endpoints.

4.2 Paper-Level Outcomes and Period Comparison

The principal endpoint comprised 41 substantive statistical issues: 37 confirmed and 4 probable. At least one was present in 22 of 44 papers (50.0%; exact binomial 95% confidence interval, 34.6% to 65.4%). There were 14, 6, 11, and 10 eligible issues in 2018, 2019, 2024, and 2025, respectively. Separately, 149 confirmed report-material issues of any substantiveness were present in 38 of 44 papers; this broader count includes minor presentation issues and is distinct from the principal endpoint. Neither proportion estimates the prevalence of inconsistencies in *JAMA* generally.

In 2018–2019, the principal endpoint included 20 issues in 19 papers, a mean of 1.05 per paper; in 2024–2025, it included 21 issues in 25 papers, a mean of 0.84. The later-minus-earlier difference was -0.21 issues per paper (95% bootstrap confidence interval, -1.02 to 0.56; exact permutation $P = 0.654$). The proportions of papers with at least one eligible issue were 10/19 (52.6%) and 12/25 (48.0%), respectively, a difference of -4.6 percentage points (95% bootstrap confidence interval, -33.7 to 25.7; $P = 1.000$). The intervals were compatible with differences in either direction.

Restricting the statistical endpoint to confirmed substantive issues yielded 16 issues in the earlier period and 21 in the later period, or 0.84 per paper in each period ($P = 1.000$). All four probable substantive issues occurred in the earlier period. The secondary endpoint contained one substantive report-internal inconsistency in the earlier period and three in the later period ($P = 0.622$). Combining both endpoints yielded mean counts of 1.11 and 0.96 ($P = 0.754$). Table 4 gives the corresponding differences and confidence intervals, Figure 2 shows the distribution of statistical counts, and Table 5 provides the paper-level data.

Table 4: Exploratory comparison of report-material outcomes between publication periods.

Outcome	2018–2019	2024–2025	Difference (95% CI)	*P*
Substantive statistical (count)	1.05	0.84	-0.21 (-1.02, 0.56)	0.654
Substantive statistical (any)	10/19	12/25	-4.6 (-33.7, 25.7)	1.000
Substantive report inconsistency (count)	0.05	0.12	0.07 (-0.09, 0.24)	0.622
Substantive report inconsistency (any)	1/19	3/25	6.7 (-9.1, 24.0)	0.622
Confirmed statistical only (count)	0.84	0.84	0.00 (-0.77, 0.74)	1.000
Confirmed statistical only (any)	8/19	12/25	5.9 (-24.4, 35.2)	0.766
Combined substantive report (count)	1.11	0.96	-0.15 (-1.02, 0.68)	0.754
Combined substantive report (any)	10/19	12/25	-4.6 (-33.7, 25.7)	1.000

Count rows show the mean number per paper; any rows show papers with at least one eligible issue. Differences are later minus earlier, in issues per paper or percentage points, respectively. Principal-endpoint eligibility includes confirmed or probable statistical issues judged substantive. The confirmed-only and combined endpoints are sensitivity analyses. Confidence intervals use 20,000 paper bootstrap resamples within periods; two-sided permutation *P* values are exact and unadjusted. No protocol–report or planning-material finding contributes.

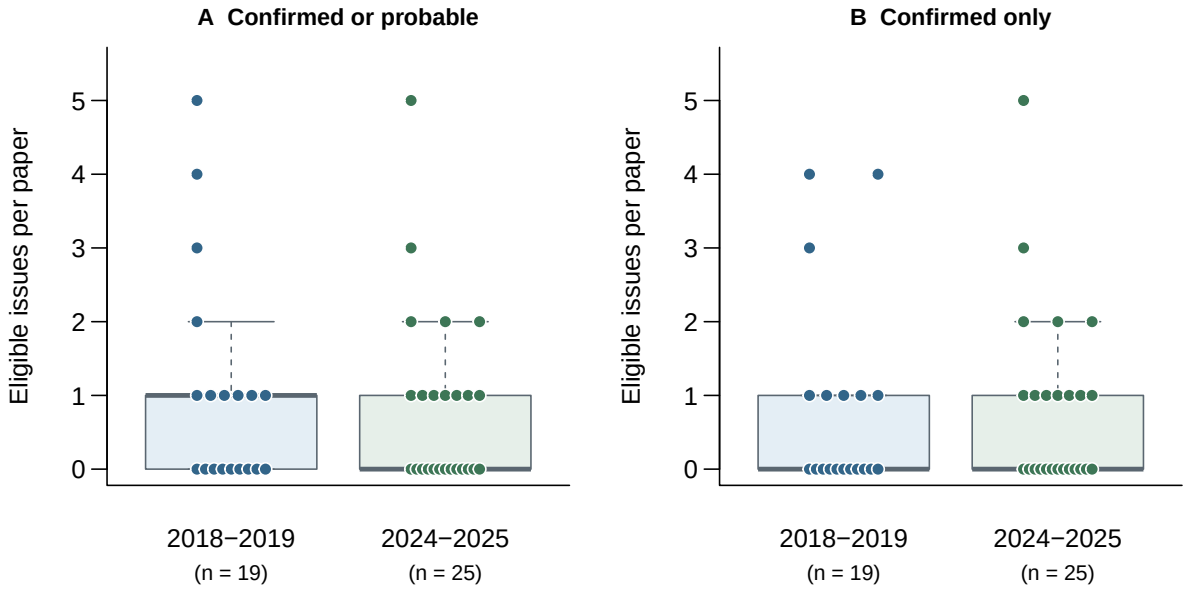


Figure 2: Substantive statistical issues per paper by publication period. Each point represents one paper, including papers with zero eligible issues. Panel A includes confirmed and probable issues; Panel B includes confirmed issues only. Boxes show the median and interquartile range, with whiskers extending to observations within 1.5 interquartile ranges. Horizontal offsets separate overlapping points.

5 Discussion

In this availability sample of 44 trials, human adjudication retained 413 of 440 candidates (93.9%). Retention included uncertain findings and protocol–report differences; the retention rate does not measure confirmation of reporting issues or workflow accuracy. The principal endpoint comprised 41 retained, confirmed or probable substantive statistical issues in report materials, distributed across 22 papers. Mean counts were 1.05 per paper in 2018–2019 and 0.84 in 2024–2025 (difference, -0.21; 95% bootstrap confidence interval, -1.02 to 0.56; exact permutation $P = 0.654$). These data did not establish a period difference, and the interval does not support a conclusion of equivalence.

TRACE-Q organizes quantitative review around relationships among reported quantities and the evidence needed to assess them. Its practical contribution is the connection between source locations, checking records, screening decisions, and the final candidate report. Saving these records can preserve work across agent contexts and allow a human reviewer to revisit the basis of a concern. Reusing extraction assets is useful only while their source identity and coverage remain clear; an optical character recognition error must be resolved against the original document before it is treated as a reporting concern. These workflow properties do not establish that every relevant relationship was recognized or every interpretation was correct.

The distinction between discovery and verification matters. La et al. assessed the recovery of planted inconsistencies and adjudicated additional findings in ophthalmic articles.[9] FIND, FLAWS, and SPOT likewise evaluate the recovery of specified targets, although their targets and documents differ.[5–7] Our adjudications concern candidates already produced by the workflow. The proportion retained describes those candidates under the applied codebook; it does not measure the probability of detecting every problem in a paper. A paper without a retained candidate therefore cannot be classified as free of reporting problems. Nor do candidate counts establish effects on a trial's principal conclusion, clinical interpretation, or subsequent evidence synthesis.

The comparison between publication periods requires similar restraint. We selected articles by availability, rather than drawing a probability sample of *JAMA* trials. Differences in clinical topic, trial design, reporting complexity, or the amount of supplementary and planning material could affect the number of relationships available to inspect and the types of candidates generated. Permutation tests provide an exploratory comparison within the assembled sample, but cannot remove selection effects or establish a causal effect of publication period. Findings should therefore be interpreted as characteristics of these article packages and this review process, rather than a trend in the quality of medical research.

Several limitations concern the review itself. This application formed part of workflow development and did not provide an independent held-out validation. Exposure to earlier cohort outputs may have influenced the review rules. One reviewer supplied the final adjudications, and we did not estimate agreement between independent reviewers. Decisions about whether context resolves a discrepancy or whether an issue materially affects extraction may depend on expertise and interpretation. The paper is also the relevant unit of analysis: several candidates can arise from the same analysis, document, or reporting practice. Counting them as independent observations would overstate the amount of evidence. The adjudicated workbooks preserve the recorded decisions, but those decisions remain subject to human reassessment.

Some screening rules also require statistical qualifications. A confidence interval and a P value can differ because they use different procedures, adjustment sets, or inferential conventions. An adjusted estimate cannot generally be checked by substituting a crude ratio, and a change from a planned analysis may be explained by a documented amendment. Display-precision thresholds and clinical rules of thumb are therefore aids to identifying a question for review. They do not establish a universal boundary between acceptable reporting and a substantive error. Dong et al.'s findings about extraction, omitted analyses, and

dependence on reported values reinforce the need to examine the entire checking process.[10] Missing participant-level data or analysis code may leave a plausible concern unresolved even when the published material has been fully inspected.

We did not compare TRACE-Q with unaided human review, a single-agent implementation, or alternative agent environments. We also did not establish savings in total review time or cost. Delegating roles may make responsibilities explicit, but agents using the same model can share assumptions and reproduce one another's errors. The arXiv study by Kim et al. found that the effects of additional agents depended on task structure and coordination.[12] A controlled comparison with matched sources, models, tools, and computational budgets is needed to identify whether TRACE-Q's division of work improves useful discovery or reduces the human burden of adjudication. Repeated runs are also needed to evaluate stability across model versions and execution settings.

Sharing the workflow and analysis code would support reuse, but access to the underlying articles, supplements, and derived evidence may depend on publisher permissions and institutional access. Reproducibility therefore requires a clear distinction between materials that can be redistributed and source documents that readers must obtain separately. A useful release should identify the exact workflow, model settings, source versions, and adjudication codebook, with sufficient source locators to reconstruct permissible checks. Future evaluation should combine independently adjudicated natural findings with known inserted errors, retain failed or incomplete reviews in the reported denominators, and measure the time required for human verification. TRACE-Q is presently an exploratory framework for organizing evidence and directing attention to specific quantitative reporting questions; its value in routine review remains to be established prospectively.

Data and Code Availability

The accompanying analysis files contain the R script, derived candidate- and paper-level tables, source metadata, and input hashes used for this draft. The workflow specification and completed adjudication workbooks are retained with the project. Journal PDFs are not redistributed with the manuscript; access to articles and supplementary documents remains subject to the applicable publisher and institutional terms. A public repository location and the scope of any public release will be specified by the authors before submission.

Use of Artificial Intelligence

The screening application used the model and workflow described in the Methods. Draft preparation additionally used GPT-6-Astra agents for writing and a GPT-6.1-sol agent for language review, with the supplied methods paper as a style reference. Computational results were generated from the final human-review workbooks using R. The authors should verify the manuscript, citations, and final disclosure before submission.

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A Paper-Level Results

Table 5: Per-paper counts derived from the final human workbooks.

Year	Paper DOI suffix	Candidates	Retained	Primary	Strict	Secondary
2018	jama.2017.21906	7	6	1	0	0
2018	jama.2018.0156	8	7	1	1	0
2018	jama.2018.0948	12	11	0	0	0
2018	jama.2018.14280	10	10	1	1	0
2018	jama.2018.14282	16	16	3	3	1
2018	jama.2018.17075	15	15	2	1	0
2018	jama.2018.18020	22	21	5	4	0
2018	jama.2018.6496	13	11	0	0	0
2018	jama.2018.8802	5	4	1	1	0
2018	jama.2018.9128	7	6	0	0	0
2019	jama.2019.0556	11	11	1	0	0
2019	jama.2019.10517	5	5	0	0	0
2019	jama.2019.12618	10	10	0	0	0
2019	jama.2019.14630	9	9	1	1	0
2019	jama.2019.14901	5	5	0	0	0
2019	jama.2019.17380	18	18	0	0	0
2019	jama.2019.2210	7	5	0	0	0
2019	jama.2019.4755	1	1	0	0	0
2019	jama.2019.7505	13	12	4	4	0
2024	jama.2024.0318	14	14	0	0	0
2024	jama.2024.0572	19	16	5	5	0
2024	jama.2024.11057	5	5	0	0	0
2024	jama.2024.12829	7	7	1	1	0
2024	jama.2024.19585	7	5	0	0	0
2024	jama.2024.2302	6	6	1	1	0
2024	jama.2024.23898	13	13	0	0	0
2024	jama.2024.25786	15	14	1	1	0
2024	jama.2024.4183	14	14	1	1	0
2024	jama.2024.6063	13	13	2	2	0
2025	jama.2024.24764	5	5	1	1	0
2025	jama.2024.27643	16	13	2	2	1
2025	jama.2025.11178	18	17	3	3	0
2025	jama.2025.15185	6	6	1	1	0
2025	jama.2025.15440	1	1	0	0	0
2025	jama.2025.16450	6	5	0	0	0
2025	jama.2025.19563	13	10	0	0	0
2025	jama.2025.19843	7	7	0	0	0
2025	jama.2025.20765	19	19	1	1	1
2025	jama.2025.24175	14	14	2	2	1
2025	jama.2025.4390	13	13	0	0	0
2025	jama.2025.7583	4	4	0	0	0
2025	jama.2025.7710	3	3	0	0	0
2025	jama.2025.9110	3	2	0	0	0
2025	jama.2025.9663	5	4	0	0	0

DOI prefix is 10.1001/. Primary = substantive confirmed/probable statistical issues; strict = substantive confirmed statistical issues; secondary = substantive report-internal inconsistencies. All three require an original report-only relationship.

B Reproducible Analysis and Interpretation of Review Fields

The analysis script reads the four completed Excel workbooks directly and uses stable candidate keys to join the original relationship codes. Blank rows in workbook indices are ignored. It checks candidate uniqueness, paper-level row totals, completion status, and required domain-specific judgments. No human label is inferred from an automated report or overwritten during analysis. A candidate's final human domain determines descriptive tabulation, while its original document relationship additionally constrains temporal eligibility. In particular, candidate `2024::jama.2024.11057::C003` was retained as a substantive report-internal inconsistency after human domain editing, but its original `P_R` relationship keeps it outside the secondary temporal endpoint. The main results include its human judgment in the descriptive domain total.

The candidate and paper analysis variables are saved in `candidate_results.csv` and `paper_results.csv`. The file `period_comparisons.csv` gives the unrounded estimates, confidence limits, and exact permutation probabilities. The source catalogue retains original domain and relationship assignments. The script also compares its permutation calculation against direct enumeration on a small example containing tied and zero counts. Its use of all 44 papers ensures that lack of an eligible issue contributes a zero count rather than causing exclusion of the paper.

Table 6: Operational interpretation of the final human review fields.

Field	Interpretation used in the analysis
Disposition	Retain, exclude, or discuss; blank means unanswered. Retention alone is not confirmation.
Statistical issue	Confirmed, probable, not an issue, or indeterminate; required for retained statistical candidates.
Substantiveness	Substantive, minor presentation, or indeterminate; separate from evidential certainty.
Downstream impact	Optional direct, possible, unlikely, or unknown impact; not an eligibility criterion or evidence of actual harm.
Internal inconsistency	Substantive, minor presentation, no inconsistency, or indeterminate; applies to report-internal and planning-material candidates.
Protocol interpretation	Optional description of a difference or ambiguity; excluded from temporal endpoints regardless of apparent importance.
Completion	`primary_done` indicates completed entry and review; it does not imply scientific confirmation.